

SELBU, Norway

WMO: 012730

Lat: 63.225N

Lon: 11.008E

Elev: 160

StdP: 99.42

Time Zone: 1.00 (EUC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.5	-16.4	-21.7	0.5	-19.1	-18.9	0.7	-15.6	12.4	-1.3	10.4	-0.2	1.0	150	0.626

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	25.7	16.8	23.4	15.6	21.3	14.8	17.8	23.7	16.5	21.8	15.4	20.3	2.9	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.2	11.0	20.7	14.1	10.2	19.2	13.1	9.6	17.7	50.7	24.1	46.8	21.9	43.7	20.5	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
9.1	7.5	6.3	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-20.3	28.6	4.1	2.2	-23.2	30.1	-25.7	31.4	-28.0	32.6	-31.0	34.2		
			-20.3	19.7	3.8	1.6	-23.1	20.8	-25.4	21.7	-27.5	22.6	-30.3	23.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.0	-3.9	-2.9	0.0	3.9	8.5	11.7	15.1	14.0	10.7	5.1	0.3	-2.6
	DBStd	7.78	5.65	5.44	4.12	3.14	3.89	3.15	3.49	2.72	2.86	3.33	4.76	5.48
	HDD10.0	2246	431	361	311	186	77	16	1	2	24	155	292	392
	HDD18.3	4868	689	594	569	433	304	199	113	137	229	409	542	650
	CDD10.0	435	0	0	0	2	31	67	160	125	46	4	0	0
	CDD18.3	16	0	0	0	0	0	1	13	2	0	0	0	0
	CDH23.3	207	0	0	0	0	8	27	148	24	1	0	0	0
	CDH26.7	44	0	0	0	0	1	3	39	1	0	0	0	0
(14) Wind	WSAvg	2.3	2.8	2.5	2.7	2.2	2.2	1.9	1.8	1.9	2.1	2.5	2.4	2.5
(15) Precipitation	PrecAvg	918	73	72	63	43	64	95	99	100	91	79	69	70
	PrecMax	1127	172	119	136	132	100	160	165	162	226	166	204	149
	PrecMin	691	7	13	29	15	39	33	37	42	23	27	3	14
	PrecStd	119	42	31	32	27	18	29	38	39	54	32	55	37
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.9	8.1	9.6	18.5	24.4	26.7	30.6	25.8	21.9	16.6	11.5	8.1
		MCWB	3.5	4.0	4.1	10.1	13.9	16.2	19.3	16.8	14.1	10.6	8.1	4.9
	2%	DB	5.2	6.1	7.6	15.0	21.7	22.6	27.3	23.5	19.4	13.8	9.1	6.4
		MCWB	2.4	2.6	3.7	8.1	13.1	14.7	18.2	15.7	13.7	9.5	6.2	3.5
	5%	DB	4.0	4.8	6.4	12.0	18.9	20.0	24.5	21.6	17.5	12.0	7.6	5.1
		MCWB	1.7	2.0	3.1	6.4	11.8	13.8	16.7	15.1	12.6	8.4	4.9	2.5
	10%	DB	2.8	3.7	5.3	9.5	16.0	17.8	22.1	19.7	15.8	10.2	6.3	4.0
		MCWB	0.9	1.3	2.2	4.9	10.3	12.5	15.8	14.4	11.7	6.9	3.9	1.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.3	4.4	5.7	10.3	15.3	16.8	20.6	18.1	15.7	12.5	8.2	5.3
		MCDB	6.0	7.3	8.0	17.8	22.5	23.3	28.6	23.2	19.8	14.1	10.7	7.4
	2%	WB	2.9	3.2	4.4	8.3	13.6	15.4	19.0	16.7	14.2	9.8	6.6	3.9
		MCDB	4.6	5.3	6.9	14.1	20.3	21.8	26.1	21.7	18.4	13.4	8.6	5.9
	5%	WB	2.0	2.2	3.5	6.6	12.1	14.1	17.4	15.8	13.1	8.6	5.2	2.7
		MCDB	3.6	4.3	5.9	11.1	18.6	19.1	23.3	20.3	16.7	11.7	7.0	4.7
	10%	WB	1.1	1.5	2.6	5.3	10.6	13.0	16.2	14.7	12.0	7.3	4.1	1.8
		MCDB	2.6	3.5	4.8	9.0	15.6	17.1	21.3	18.8	15.4	9.8	6.2	3.5
(35) Mean Daily Temperature Range	5% DB	MDBR	6.0	6.7	8.0	9.3	9.7	8.6	9.4	9.3	8.3	7.0	5.6	5.8
		MCDBR	5.8	6.3	8.1	13.6	15.2	13.2	14.5	12.6	11.4	9.3	7.1	6.4
		MCWBR	4.2	4.3	5.1	7.5	7.9	6.8	6.9	6.4	6.6	6.0	4.9	4.9
	5% WB	MCDBR	5.2	6.0	7.1	12.0	14.5	12.0	13.4	11.0	10.7	8.4	6.9	6.2
		MCWBR	3.9	4.4	4.8	6.9	7.8	6.6	6.8	6.1	6.5	5.6	5.1	4.8
(40) Clear-Sky Solar Irradiance	taub		0.217	0.277	0.309	0.338	0.346	0.345	0.355	0.343	0.318	0.298	0.230	0.356
	taud		2.033	2.213	2.235	2.302	2.414	2.466	2.458	2.487	2.541	2.449	2.114	2.701
	Ebn at Noon		411	661	784	842	869	877	855	836	791	651	391	240
	Edh at Noon		30	60	87	102	100	97	96	85	66	49	27	18
(44) All-Sky Solar Radiation	RadAvg		0.17	0.73	2.05	3.85	4.65	4.79	4.62	3.55	2.16	0.96	0.26	0.09
	RadStd		0.03	0.10	0.26	0.29	0.30	0.47	0.48	0.40	0.22	0.13	0.04	0.01

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-142	-178	N/A	N/A

Nomenclature: See separate page